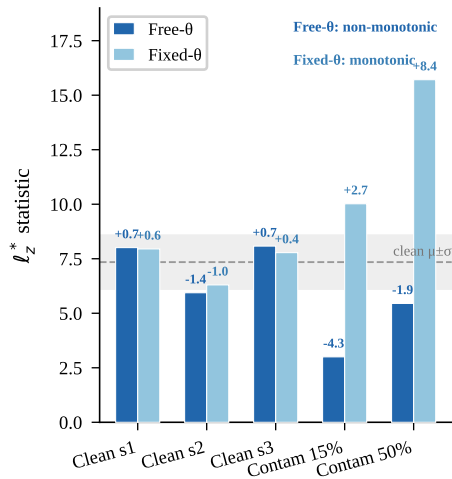
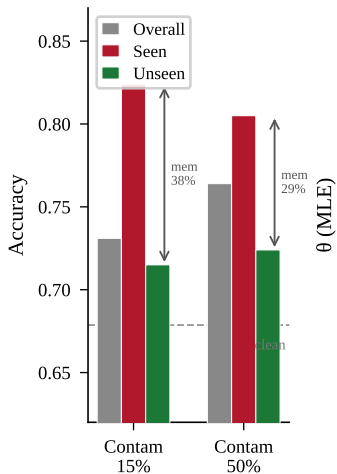


(a) ℓ_z^* under Free- θ vs Fixed- θ 

(b) Memorization Fingerprint

(c) θ -Shift Mechanism